ECE 537: Advanced Data Analytics

Exercise #2:

Bayesian Classification: A Prior Probabilities, Loss Functions, and Impacts of Outliers on Decision Boundaries

Bayes classifiers provide decision boundaries that minimize misclassification risk. These classifiers are described in Chapter 2 of "Pattern Classification" by R. Duda *et al* and can be found in most other texts on pattern classification and statistical data analysis under the "Bayes Classifier" label.

Additional background information relevant to the exercises can be found on the ECE 537 Brightspace webite under the General Reference Materials tab.

Tasks:

- Use Matlab's Live Script feature to create a live script that does (and documents) the following work:

1. Reload the Class A, B and C data from Exercise 1 and generated the Bayes decision boundaries for equal a priori class probabilities.

2. Assess the degree to which changes to the means and covariance of the data classes produces changes the nature of the generated inter-class Bayes decision boundaries.

3. Provide two example cases where relatively minor variations in the class A, B, and C means and/or covariances lead to quite significant changes in the 3-D plotted multi-class decision boundary. Provide two separate plots showing differences in the produced 3-D boundaries.

   Describe why small changes in the class means and/or covariances can lead to large changes in the decision boundary's shape and form,

4. Assess how these decision boundaries change with changes to the three class's a priori probabilities.

5. Assess how these decision boundaries change with changes to the classes $\lambda_{AA}, \lambda_{AB}, \lambda_{AC}, \lambda_{BB}, \lambda_{BC}$, and, λ_{CC} loss functions, assuming $\lambda_{AB} = \lambda_{BA}$, $\lambda_{AC} = \lambda_{CA}$ and $\lambda_{BC} = \lambda_{CB}$.

6. Now introduce outlier data points to one of the classes at a 10, 50, and 100 standard deviations from that class's mean and regenerate the Bayes class boundaries. Note measuring how far a point is from the mean of a class in terms of that class's standard deviations is defined be a measurement of that point's *Mahalanobis distance* from the mean.

    - Discuss the impacts that such *single* outlier data points can have on the resulting nature of the inter-class Bayes decision boundaries.

7. For real-world data analysis problems,

    (a) Could the nature of the class decision boundaries effective be driven by only a few data points within the data?

    (b) Is it correct (or appropriate) to have decision boundaries that exhibit high sensitivities to only a small number of data samples?

    (c) How would you detect this?

    (d) How would you resolve this?